

Closit: Clipped-Regime Posit Quantization for Edge-Friendly Vision Transformers

**Sungsoo Han, Dahun Choi
and Hyun Kim**

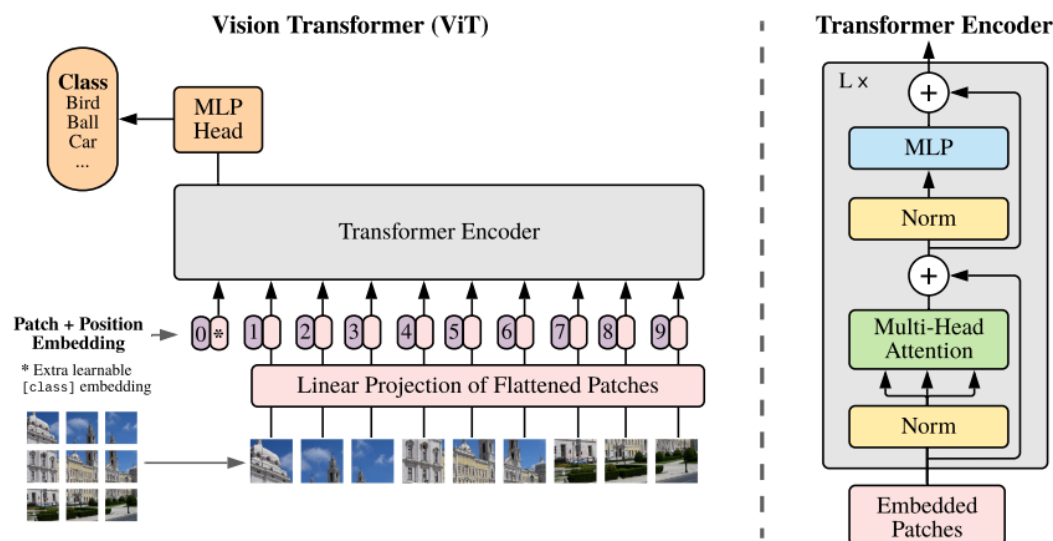
***Seoul National University of Science and Technology
Electrical and Information Engineering***

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Motivation - 1

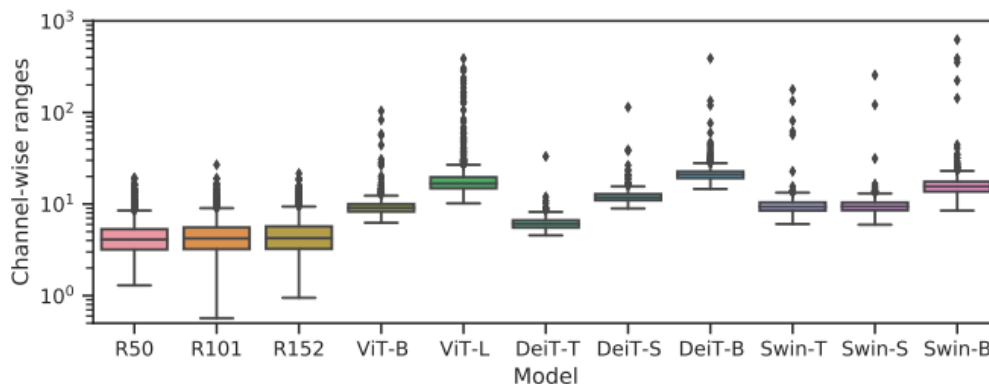
- **Quantization in Vision Transformers**
 - **Vision Transformers (ViTs)**
 - Achieve excellent accuracy in computer vision tasks
 - High inference cost due to model size and complexity
 - Difficult to deploy on resource-constrained edge devices



<Overview of Vision Transformers[1]>

Motivation - 2

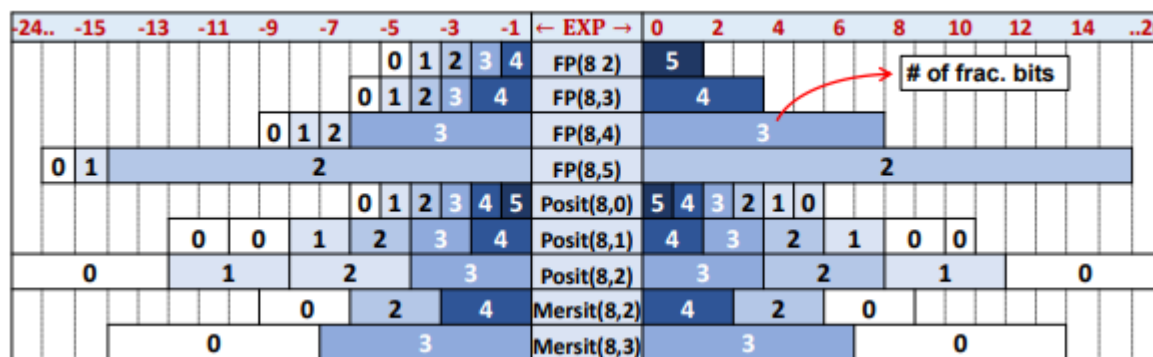
- **Quantization in Vision Transformers**
 - **The Outlier Challenge in ViTs**
 - More pronounced outliers compared to traditional Deep Neural Networks (DNNs)
 - Integer (INT) quantization isn't robust to outliers
 - **Emergence of New Approaches for Data Formats**
 - Wider range formats (e.g., FP, Posit) are emerging as solutions



<Data range from various models[2]>

Motivation - 3

- Posit
 - Characteristic of Posit
 - Provides dynamic precision with wider range
 - Decoding step introduces significant hardware overhead
 - Inherent Issue of Data Format
 - The format is not optimized for the unique statistical properties of DNN data distribution



<Range and precision of various 8-bit data formats[3]>

Contribution

- **Closit: Clipped-Regime Posit**
 - **Goal of Closit**
 - Hardware-efficient solution~(1) with appropriate range~(2), not just wider range, and dynamic precision allocation to critical ranges~(3)
 - **Regime Clipping~(1),(2)**
 - Clips regime range for appropriate range coverage
 - Simplifies decoding process, reducing area and power of Posit decoder by 46.2% and 61.9% respectively
 - **Regime Remap~(3)**
 - Adaptively remaps dynamic precision to DNN data distribution, showing 0.01% accuracy drop in ViT-B

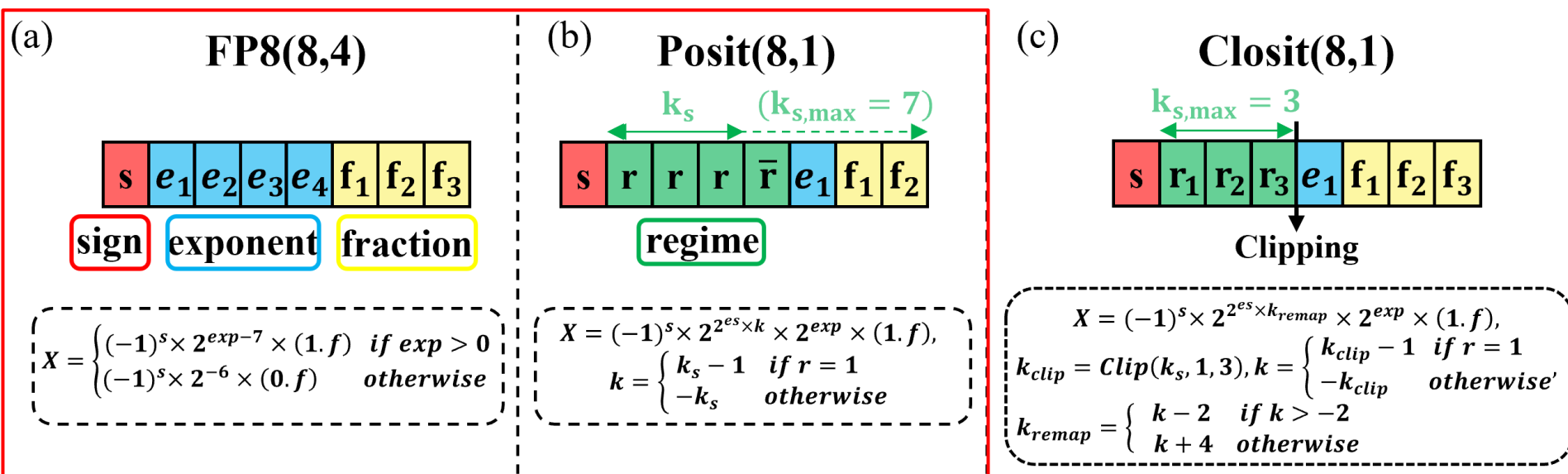
Backgrounds

- **Floating Point**

- Consists of sign, exponent (exp), and fraction(frac), providing fixed range and precision determined by exponent bits

- **Posit**

- Introduces regime for wider range representation, with precision that changes dynamically according to regime



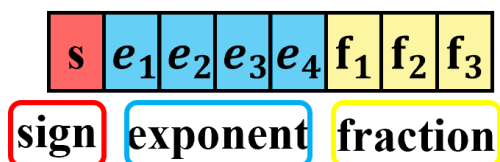
<Bit-field configurations for (a) FP(8,4), (b) Posit(8,1), (c) Closit(8,1)>

Proposed Method - 1

- **Regime Clipping**
 - Significantly reduces decoding overhead by clipping regime range to a maximum of 3-bit
 - Reduces 6-bit Leading Zero Detector (LZD) to 2-bit LZD
 - Simplifying logic for determining exp and frac fields

(a)

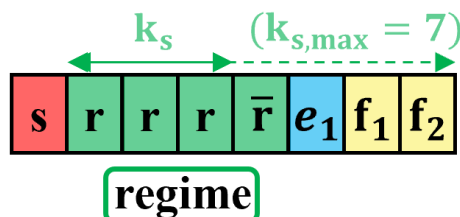
FP8(8,4)



$$X = \begin{cases} (-1)^s \times 2^{exp-7} \times (1.f) & \text{if } exp > 0 \\ (-1)^s \times 2^{-6} \times (0.f) & \text{otherwise} \end{cases}$$

(b)

Posit(8,1)

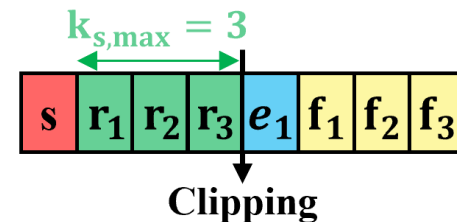


$$X = (-1)^s \times 2^{2^{es} \times k} \times 2^{exp} \times (1.f),$$

$$k = \begin{cases} k_s - 1 & \text{if } r = 1 \\ -k_s & \text{otherwise} \end{cases}$$

(c)

Closit(8,1)



$$X = (-1)^s \times 2^{2^{es} \times k_{remap}} \times 2^{exp} \times (1.f),$$

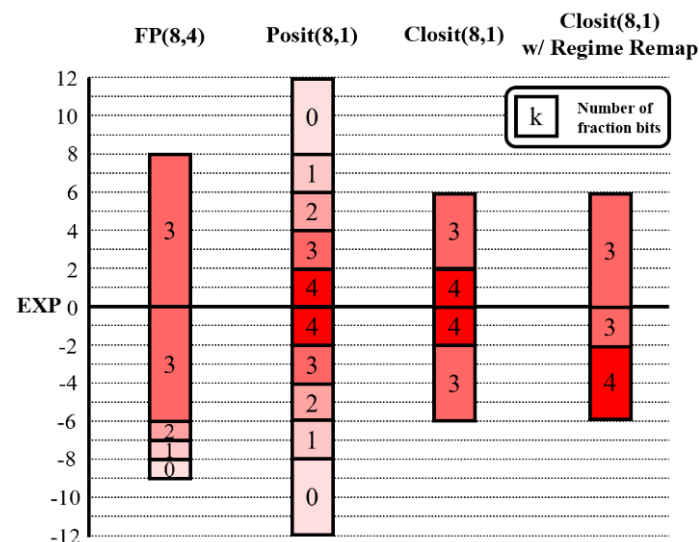
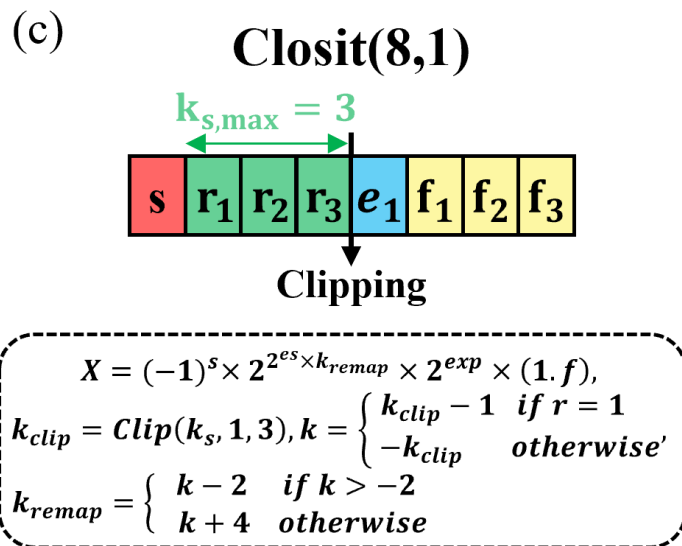
$$k_{clip} = \text{Clip}(k_s, 1, 3), k = \begin{cases} k_{clip} - 1 & \text{if } r = 1 \\ -k_{clip} & \text{otherwise} \end{cases}$$

$$k_{remap} = \begin{cases} k - 2 & \text{if } k > -2 \\ k + 4 & \text{otherwise} \end{cases}$$

<Bit-field configurations for (a) FP(8,4), (b) Posit(8,1), (c) Closit(8,1)>

Proposed Method - 2

- **Regime Remap**
 - Remaps regime value to adaptively allocate dynamic precision according to DNN data distribution
 - Conventional Posit allocates high precision around the value 1
 - Uses MSE-based remapping optimized for DNN data distribution



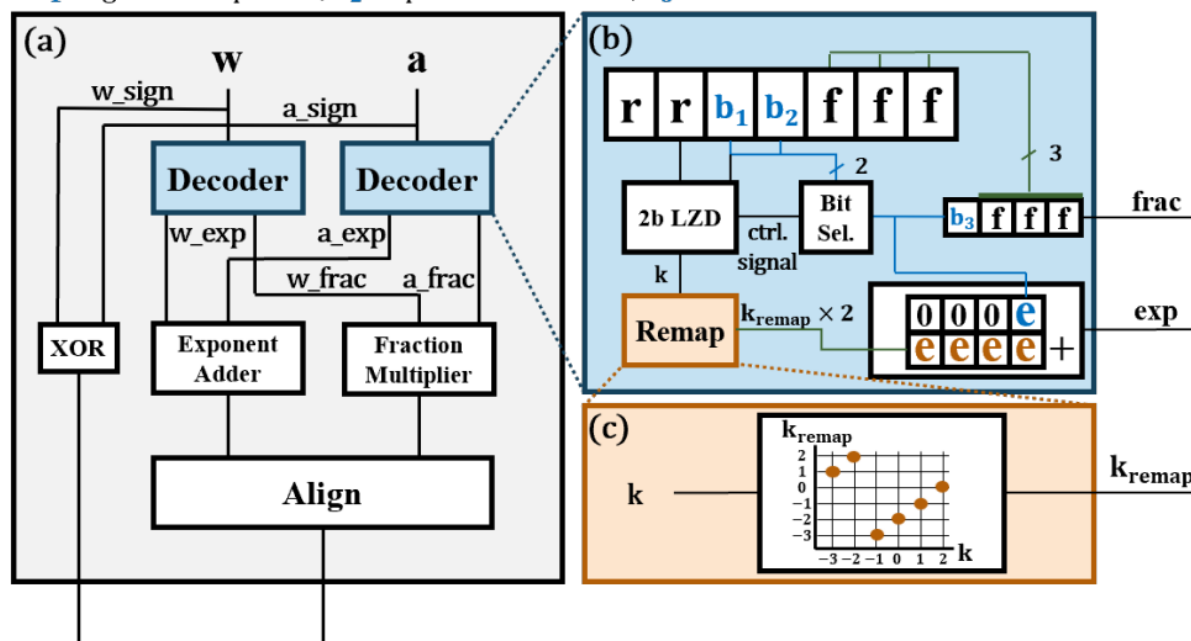
<Bit-field configurations for (c) Closit(8,1)>

< Range and precision of 8-bit data formats >

Proposed Method - 3

- **Decoding & Multiply Scheme of Closit**
 - Simplifies LZD module and exp/frac field decision process during decoding
 - Introduces minimal overhead occurs from the remapping

* b_1 : regime or exponent, b_2 : exponent or fraction, b_3 : fraction or '0'



<Decoding and multiply scheme of closit(8,1)>

Experimental Results - 1

- **Experimental Setup**
 - **Dataset**
 - ImageNet-1k
 - Calibration for 32 images
 - **Model**
 - ViT (Tiny, Small, Base)
 - DeiT (Tiny, Small, Base)
 - **Data Format**
 - INT8
 - FP(8,4)
 - Posit(8,1)
 - **Hardware**
 - RTL designs were implemented in Verilog and synthesized using Synopsys Design Compiler with a 28nm technology library

Experimental Results - 2

• Accuracy Analysis

- Closit(8,1) achieves the best accuracy across all results
- **INT shows larger accuracy drop** compared to other formats
- FP(8,4) and Posit(8,1) show comparable performance
- **Closit(8,1) achieves only 0.01% accuracy drop** compared to baseline in ViT-B

TABLE I
COMPARISON OF TOP-1 ACCURACY AND ACCURACY DROP (%) ON IMAGENET FOR VARIOUS DATA FORMATS

Model	Data Format	Precision (W/A)	Accuracy (%)	Drop (%)	Model	Data Format	Precision (W/A)	Accuracy (%)	Drop (%)
DeiT-T	FP32	32/32	72.17	-	ViT-T	FP32	32/32	75.46	-
	INT8	8/8	70.92	1.25		INT8	8/8	70.65	4.81
	FP(8,4)	8/8	71.71	0.46		FP(8,4)	8/8	74.77	0.69
	Posit(8,1)	8/8	71.65	0.52		Posit(8,1)	8/8	74.84	0.62
	Closit(8,1) (ours)	8/8	71.93	0.24		Closit(8,1) (ours)	8/8	75.08	0.38
DeiT-S	FP32	32/32	79.86	-	ViT-S	FP32	32/32	81.38	-
	INT8	8/8	79.12	0.74		INT8	8/8	77.88	3.50
	FP(8,4)	8/8	79.70	0.16		FP(8,4)	8/8	81.11	0.27
	Posit(8,1)	8/8	79.61	0.25		Posit(8,1)	8/8	81.18	0.20
	Closit(8,1) (ours)	8/8	79.82	0.04		Closit(8,1) (ours)	8/8	81.29	0.09
DeiT-B	FP32	32/32	81.98	-	ViT-B	FP32	32/32	85.10	-
	INT8	8/8	81.24	0.74		INT8	8/8	84.40	0.70
	FP(8,4)	8/8	81.76	0.22		FP(8,4)	8/8	84.25	0.85
	Posit(8,1)	8/8	81.76	0.22		Posit(8,1)	8/8	84.86	0.24
	Closit(8,1) (ours)	8/8	81.86	0.12		Closit(8,1) (ours)	8/8	85.09	0.01

Experimental Results - 3

- Hardware Analysis
 - Closit(8,1) reduces power and area by 61.9% and 46.2% respectively compared to conventional Posit decoder
 - Remap technique incurs minimal 6.3% area overhead ($22.0\mu\text{m}^2 \blacktriangleright 23.4\mu\text{m}^2$)

TABLE II
AREA AND POWER RESULTS OF DECODER AND MULTIPLIER
IMPLEMENTATIONS WITH VARIOUS DATA FORMATS

Module Type	Data Format	Power (μW)	Area (μm^2)
Decoder	Posit(8,1)	11.31	43.52
	Closit(8,1)	4.31	23.4
Multiplier	FP(8,4)	30.83	114.31
	Posit(8,1)	31.45	101.67
	Closit(8,1)	30.01	97.22

Conclusion

- **Proposed Closit: A Posit-based hardware-efficient data format**
- **Significantly simplifies decoding process using regime clipping technique**
- **Achieves excellent performance by adaptively handling DNN data distribution via regime remap**
- **Has scalable potential by extending regime clipping and remap to adaptively allocate parameter values such as remapping factor, maximum clipping length**
- **However, occupies larger power and area compared to FP**

Thank You!

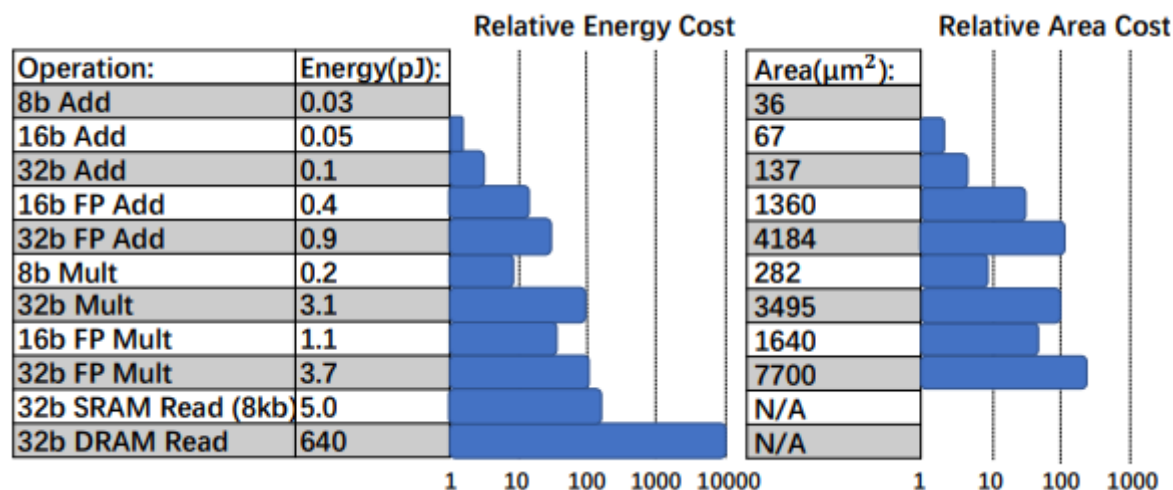
Appendix

- **[1] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., ... & Houlsby, N. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929.**
- **[2] Lin, Y., Zhang, T., Sun, P., Li, Z., & Zhou, S. (2021). Fq-vit: Post-training quantization for fully quantized vision transformer. arXiv preprint arXiv:2111.13824.**
- **[3] Ho, N. D., Jeong, G., Kang, C. M., Choi, S., & Chang, I. J. (2024, June). MERSIT: A Hardware-Efficient 8-bit Data Format with Enhanced Post-Training Quantization DNN Accuracy. In Proceedings of the 61st ACM/IEEE Design Automation Conference (pp. 1-6).**

Q & A

Appendix

- Energy, area about various data format operation



- Gholami, A., Kim, S., Dong, Z., Yao, Z., Mahoney, M. W., & Keutzer, K. (2022). A survey of quantization methods for efficient neural network inference. In *Low-power computer vision* (pp. 291-326). Chapman and Hall/CRC.

Appendix

- Without Regime Remap table

8/8	FP32	Int8 w sf	FP E4M3 w sf	Posit 8/8 w/o sf	Closit w sf (ours)	Closit + Bias w sf (ours)
vit_tiny_patch16_224	75.458	70.65	74.77	74.84	74.83	75.08
vit_small_patch16_224	81.38	77.88	81.11	81.18	81.25	81.29
vit_base_patch16_224	85.104	84.40	84.25	84.862	85.05	85.09
deit_tiny_patch16_224	72.166	70.92	71.71	71.65	71.72	71.93
deit_small_patch16_224	79.856	79.12	79.70	79.61	79.75	79.82
deit_base_patch16_224	81.98	81.24	81.76	81.76	81.82	81.86